

RECOURSE

Adversarial Claims Adjudication

Five AI agents put every disputed insurance claim on trial — and the debate *becomes* the legally-defensible audit trail.

recourseband.duckdns.org

Live in production

5 Band agents

THE PROBLEM

A denied claim. A dispute. And no record of how the call was made.

- Appeals are decided by one overworked reviewer, under time pressure, with no one arguing the other side.
- The reasoning lives in someone's head — not on paper. Regulators and customers can't see *why*.
- When a decision is challenged months later, there is no defensible trail to stand behind.

1

reviewer per appeal — no adversary, no second lens

Hours

of manual policy cross-referencing per case

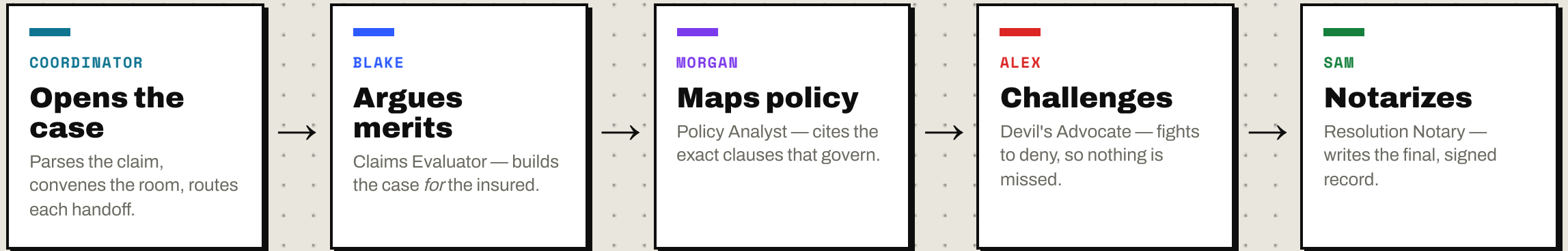
Zero

structured audit trail of the reasoning

THE SOLUTION

Five specialists put every claim on trial.

A Coordinator convenes four specialist agents in a single Band room. Each owns one job. They argue in the open, hand off to one another, and converge on a resolution — with a human officer holding the final word.



How a verdict is reached.

01

Intake

Coordinator opens the case & loads the policy.

02

Argue

Blake makes the case for paying the claim.

03

Policy

Morgan cites the governing clauses (RAG).

04

Challenge

Alex attacks the weakest points to deny.

05

Compile

Coordinator assembles the full record + math.

06

Notarize

Sam writes the signed, reasoned resolution.

07

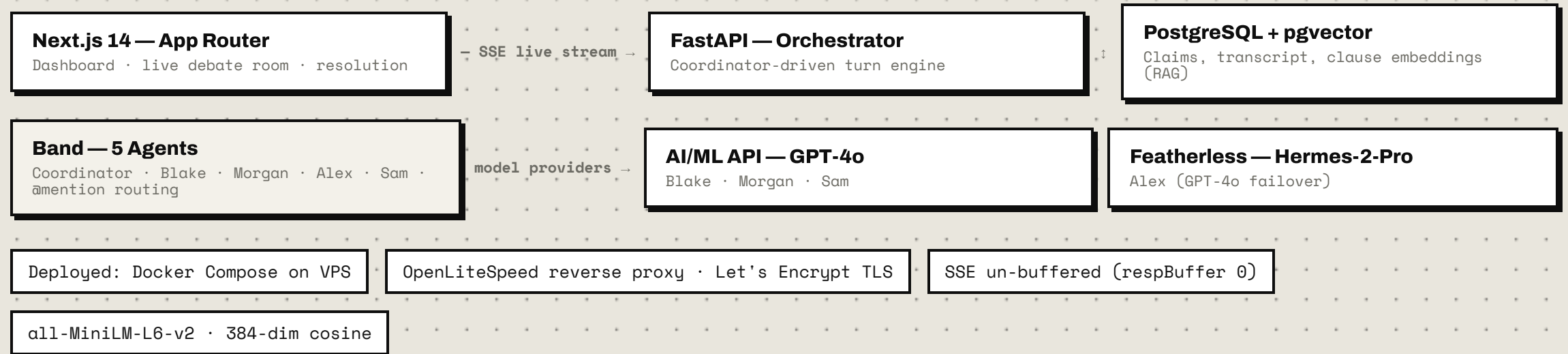
Sign-off

A human officer approves — or overrides.

THE KEY IDEA

The conversation *is* the audit trail. Every message is persisted, ordered, and hashed.

One room. A streaming spine. A durable record.



THE VERDICT

A signed result — built to survive a regulator.

- Sam issues a **binding resolution**: decision, payable amount, reasoning.
- Every clause is **cited by number** — §7.3, §12.1 — never paraphrased.
- The full transcript is **SHA-256 hashed** — tamper-evident on the record.
- Payout is deterministic: $\text{payable} = \text{claim} - \text{deductible}$, clamped at zero.
- A human officer can **approve or override** — accountability stays human.

Debate complete. Compiling the full record and routing it to @Sam for the binding resolution. (Claim \$12,500.00 · deductible \$500.00 · payable in full \$12,000.00.)

SAM RESOLUTION NOTARY 03:35:42

@Coordinator ---

DECISION: APPROVED

APPROVED AMOUNT: \$12,000.00

LEGAL REASONING:

The claim is approved based on the exception in §12.1, which allows for coverage of mechanical failures directly caused by a covered collision event. The **mechanic report** **BM-AUTO-2024-089** and **police report** **FHP-2024-10153** confirm the engine seizure was due to collision impact, not a pre-existing issue, thus negating the application of §7.3.

DEBATE SUMMARY:

- ▶ **Blake:** The coverage analysis was unclear, with a 60% confidence level, due to potential applicability of §7.3.
- ▶ **Morgan:** Asserted that §7.3 does not apply due to the exception in §12.1, supporting coverage.
- ▶ **Alex:** Argued that the denial overlooked the exception in §12.1, advocating for claim approval based on the evidence.

CONFIDENCE: HIGH

RECOMMENDATION TO CLAIMS OFFICER: Approve the claim for \$12,000.00, as the evidence and policy language support coverage under the collision clause.

Sam's signed resolution = decision, amount, legal reasoning & confidence

Real agents argue it out, in the open.

clearly supports coverage.

Handing off to **@Alex** for further review from the insured's perspective.



ALEX

DEVIL'S ADVOCATE 03:35:35

Wait. The denial hinges on **\$7.3**, but the exception in **\$12.1** is pivotal here. The **mechanic report** **BM-AUTO-2024-089** explicitly states the engine seized due to collision impact, not a pre-existing issue. This aligns with **\$12.1**, which allows coverage for failures directly caused by a collision. The **police report** **FHP-2024-10153** and witness statement confirm the vehicle was operational until the collision, contradicting any assertion of prior mechanical failure.

Moreover, the adjuster's conclusion lacks support from the evidence. **The photos** (**CLM-2024-04471-imgs**) show damage consistent with a high-impact collision, reinforcing the mechanic's findings. The policy's language is clear: when a collision causes a mechanical failure, coverage applies. The denial overlooks this exception, misinterpreting the policy terms to the detriment of the insured.

Given these points, the denial should be overturned, and the claim approved under the collision coverage, less the deductible. The evidence and policy language unequivocally support this outcome.

RECURSE

ADVERSARIAL CLAIMS ADJUDICATION

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The docket — recourseband.duckdns.org

Adjudication room — agents cite clauses (\$7.3, \$12.1), @mention each other, and stream live over SSE

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How we use Band — and the partner stack.

BAND COORDINATION

- 5 agents in one room via the **Agent API**
- Visibility model respected — agents act on **@mentions**
- Structured **agent-to-agent handoffs** drive the turn order
- The room transcript is the system of record

RETRIEVAL & REASONING

- Policy clauses embedded with **MiniLM** (384-dim)
- **pgvector** cosine search grounds every citation
- GPT-4o for Blake · Morgan · Sam
- Hermes-2-Pro (Featherless) for Alex, GPT-4o failover

Band SDK

AI/ML API

Featherless AI

FastAPI · async SQLAlchemy

Next.js 14 · TypeScript · Tailwind

Postgres 16 · pgvector

Docker · OpenLiteSpeed

RECOURSE

Every claim deserves a fair fight.

Adversarial adjudication, on the record — and live in production today.

recourseband.duckdns.org

Built for the Band of Agents Hackathon